

Intro to Instruction Rhetorizing

1) Rhetorizing Personal Assistant $\leftarrow$ Instruction Dataset $\leftarrow$ Instruction Rhetorizing

in this chapter, we will fine-tune the pretrained LLM to follow instructions

① Pretrained LLMs are good at text completion.

② They struggle with following instructions

$\rightarrow$ fix the grammar in this text

$\rightarrow$ convert this text into passive voice

③ Instructions $\leftarrow$ ^{stages} ^{inputs for} LLM $\rightarrow$ Desired Response ^{Goal for LLM is to generate a desired response}

convert 45 km/s to meters $\rightarrow$ 45 km/s is 45000 meters

Provide a synonym for 'bright' $\rightarrow$ A synonym for 'bright' is 'radiant'

Edit the following

sentence to remove $\rightarrow$ The artist composed the song.

all passive voice! $\rightarrow$ * Training on a dataset where the input-output pairs are explicitly provided. That's "Supervised Instruction Rhetorizing"

"The song was composed by the artist."

④ Instruction Fine-tuning workflow!
(we start w/ downloading, inspecting, and preparing the dataset that we will use to fine-tune model)

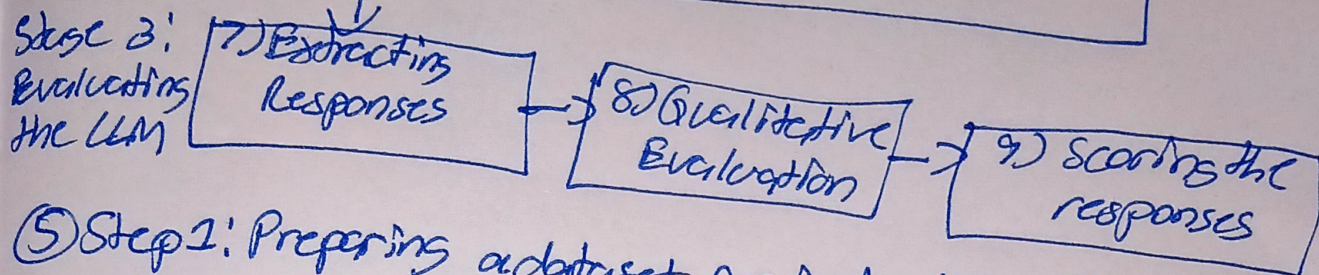
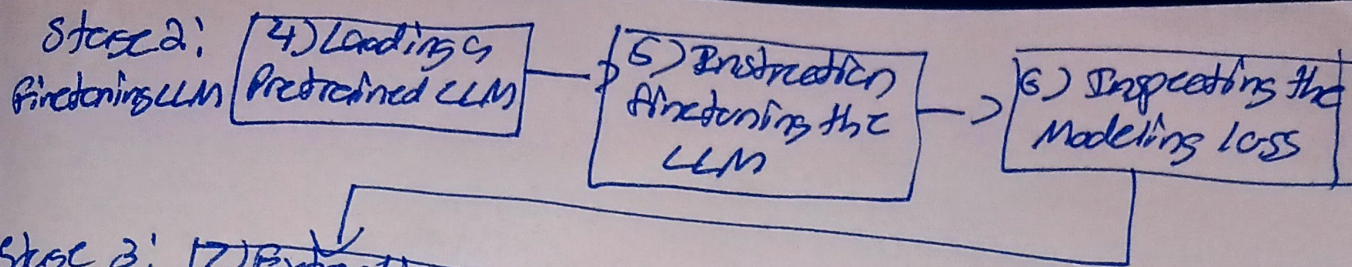
Step 1: prepare dataset

1) Dataset download and formatting

2) Batching the Dataset

3) Creating Data loaders

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⑤ Step 1: Preparing a dataset for instruction fine-tuning
 * How to format input-output pairs:

Example data point:

{
 "instruction": "Edit the following sentence for grammar.",
 "input": "He go to the park every day",
 "output": "He goes to the park every day"
 }
 ↓ Apply Alpaca prompt style template (diff. example)
 ↓ Apply Phi-3 prompt style template

Below is an instruction that describes a task. Write a response that appropriately completes the request.

Instruction:

Identify the correct spelling of the following word.

Input:

~~Occassion~~

Response:

The correct spelling is 'Occassion'.

<USER>

Identify the correct spellings of the following word: ~~Occassion~~

<ASSISTANT>

The correct spelling is 'Occassion'

↳ Instruction + input (fused)

↳ Phi-3 prompt style

* Partitioning dataset (train-test-val)
 85% 10% 5%

() Alpaca prompt style
 (More common)

⑥ Organizing data into training batches.
 (Next RAAGB)